

Lesson 4: The resistance of a wire

Electric Circuits · Year 10 · 50 minutes · AQA RP 15 / RP 3

Why this lesson is here

A longer wire gives moving charges more material to pass through. The test must keep its material, diameter and temperature the same.

Learning objective

Plan, collect and graph resistance against wire length while controlling temperature.

Core teaching point

Change the length between the crocodile clips. Measure V and I, then calculate $R = V/I$. Switch off between readings to limit heating.

Equipment and preparation

Low-voltage dc supply, switch, ammeter, voltmeter, constantan/nichrome wire, ruler, clips, leads; existing Resistance worksheet pp. 1-4.

Lesson sequence

Time	Phase	Teacher move and evidence
0-6	Retrieve	Ask where the two meters go and how R is found.
6-14	Plan and safety	Agree variables and low-voltage, switch-off routine.
14-34	Required practical A	Take ten lengths, record V/I quickly, repeat anomalies.
34-44	Process	Calculate R and begin graph; finish if needed next lesson.
44-50	Check	One-sentence trend and one specific improvement.

Workbook task guidance

Task 1: Label independent, dependent and three control variables. State why temperature matters.

Expected: Independent: length; dependent: calculated R; controls: material, diameter, temperature. Heating changes R.

Task 2: Draw the circuit: supply, switch, ammeter and test wire in series; voltmeter across only the measured wire length.

Expected: Ammeter and switch in series; voltmeter connected between crocodile clips across measured wire section.

Task 3: Collect 10-100 cm readings in the table. Calculate resistance at each length. Repeat any result that does not fit the pattern.

Expected: Values depend on apparatus; R should broadly increase with length. Verify all V/I calculations and repeat a clear anomaly.

Task 4: Plot R (y) against length (x), draw a best-fit line and write a conclusion supported by two readings.

Expected: Length on x, R on y, labelled units; positive trend, roughly proportional for constant wire and temperature.

Worked example

At 40 cm, $V = 0.80 \text{ V}$ and $I = 0.40 \text{ A}$. $R = 0.80/0.40 = 2.0 \text{ ohm}$.

AQA-style checkpoint

[6 marks] Describe how you would investigate how the resistance of a wire depends on its length. Include the circuit, measurements, controls, safety and processing.

Mark guidance: Award up to six: correct circuit including ammeter in series and voltmeter across wire; change clip separation and measure length; record V and I; calculate $R = V/I$; same material/diameter; low p.d. and switch off to hold temperature; repeat/analyse and plot R against length.

Anticipated misconception

Do not say 'measure resistance with a voltmeter': resistance is calculated from V and I here.

Assessment and next step

Collect one sentence of reasoning and one complete calculation or graph label. Start the next lesson by reteaching the most common error; students who are secure should tackle a changed numerical context.

Practical hand-off

Use the workbook practical record for data and graphing. The existing full AQA required-practical worksheet supplies detailed method, circuit diagrams and additional analysis. Use low voltage and local lab safety rules.